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Hydraulic fracturing to enhance injectivity and storage capacity of CO_2 storage reservoirs: Benefits and risks

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ABSTRACT

Several potential CO_2 storage reservoirs have been found to have insufficient porosity and permeability to support cost effective commercial-scale injection. As a result, the use of hydraulic fracturing to enhance injectivity and storage capacity of CO_2 storage reservoirs was explored. Previous modeling studies indicate that fracturing can increase storage capacity by modest to significant amounts (10%–35%), depending on model assumptions. Simulations completed as part of this study confirm that for a range of horizontal well lengths, number of fractures, fracture geometries, and fracture properties, injectivity is improved and capacity increases by 13%–71% over the base case unfractured vertical well. Intuitively, increasing the well length and number of fractures had a corresponding impact on increased capacity. Fracture area (i.e., the fracture height multiplied by its width) was an important parameter for increasing capacity but the specific geometry (e.g., the ratio of height-to-width) was unimportant. The most important aspect that affected capacity was the ability of a fracture to connect high permeability horizontal zones in the reservoir. This would allow a single well to access both layers, thereby maximizing total storage capacity of the reservoir while likely leading to an overall increase of the CO_2 footprint, which is an important consideration for Class VI UIC permitting. The results of this work demonstrate that hydraulic fracturing is an attractive option to consider when faced with an underperforming geologic carbon storage site that is at risk of causing a project to fail.

1. Introduction

Carbon capture, utilization, and storage (CCUS) technologies are gaining international support for their potential to address climate change impacts associated with the atmospheric release of anthropogenic carbon dioxide (CO_2). Globally, primary CO_2 emission sources continue to be fossil fuel power production facilities; however, other large industrial sources, such as ethanol plants, are also receiving significant attention for near-term deployment of CCUS (Greenberg et al., 2017). Domestically, the U.S. Department of Energy (DOE) has invested millions of dollars over the last decade on a number of pilot-scale projects (e.g., one to three million tonnes) sited throughout the country to promote the development of CCUS technologies and demonstrate safe and secure storage in deep geologic reservoirs (DOE, 2015). The primary focus of these efforts has been on deep saline aquifers, which represent as must as 91 %–98 % of the total domestic storage capacity (DOE, 2015).

The selection and siting of a subsurface reservoir for geologic carbon storage is based on many operational factors, such as proximity to a CO_2 source, rights of way for CO_2 pipelines, access to surface facilities, securing pore space rights, etc. However, in order for storage in deep saline aquifers to be viable for a particular power or industrial plant, the potential storage reservoir must have the capacity (hereby referred to as storage capacity) to accept the quantities of CO_2 that would be emitted over the duration of the plant's lifetime and at rates equal to its maximum CO_2 production rate (hereby referred to as injectivity). Due to the high costs associated with CO_2 transport, reservoirs are often sited in close proximity to CO_2 emission sources. As a result, the choice of available storage reservoirs may be limited to those with less than ideal properties in terms of injectivity and storage capacity. In addition, geochemical reactions that occur within saline aquifers after CO_2 injection, such as mineral dissolution/precipitation, could negatively impact the injectivity and storage capacity of deep saline aquifers (Cantucci et al., 2015; Xu et al., 2007; Izgec et al., 2008a; Shiraki and Dunn, 2000).

Development of geologic carbon storage projects builds upon decades of experience altering the subsurface for fossil fuel production. Technological advances in hydraulic fracturing or "fracking" have resulted in its ubiquitous application for oil and gas production. Directional steering allows operators to drill horizontal wells that can

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track narrow zones of resource-bearing rock for distances of two miles or more. Once drilling is complete, the horizontal wells are stimulated via multi-stage hydraulic fracturing operations, where millions of gallons of water containing a mixture of chemicals and proppants are pumped underground at sufficiently high pressures to fracture the low permeability reservoir rock. These newly developed fractures provide the pathways needed to allow the oil and/or gas to flow from the formation to the well, where it is then extracted at the surface. Today, nearly all natural gas wells drilled in the United States use this approach (EIA, 2018).

In 2010, the U.S. Environmental Protection Agency (EPA) finalized the regulations for injecting CO_2 into geologic storage formations (EPA, 2010). These regulations, known as the Class VI rule, created a framework for managing geologic sequestration projects while maintaining the goal of protecting underground sources of drinking water (USDW). The regulations include strict requirements for site characterization, CO_2 injection well construction, injection operations, site monitoring, financial liability, and record keeping/reporting. However, the Class VI requirements do not preclude the use of hydraulic fracturing, making it a potentially viable option for enhancing the injectivity and capacity of geologic carbon storage reservoirs.

The objective of this research was to assess the feasibility of using hydraulic fracturing to improve the performance of a low permeability CO_2 storage reservoir. Specifically, new simulations were performed using the FutureGen 2.0 CCUS project as a test case. Results are presented in terms of increased injectivity due to the creation of high permeability fractures in the reservoir, and provide, for the first time, estimates of improved performance for a Class VI permitted commercial-scale storage reservoir.

1.1. Field cases where injectivity was less than expected, displaying a need for permeability enhancement

This section highlights three CCUS field projects that experienced deployment challenges due to less than optimal storage properties at their proposed sites. Lack of sufficient permeability led to the abandonment of one project, resulted in a complex and costly CO_2 injection design for the second project, and for the third project, created doubt about the ability of the reservoir to support commercial-scale CCUS. Each of these projects may have benefitted from technologies that increase injectivity (e.g., hydraulic fracturing), which if successfully deployed, could have had a positive impact on the outcome of the projects.

1.1.1. Burger well

An primary objective of the Midwest Regional Carbon Sequestration Partnership (MRCSP) Phase 2 project was to develop CO_2 sequestration technology by drilling a deep test well and completing a 3000 metric ton (t) pilot-scale CO_2 injection at the site of the R.E. Burger power plant in Belmont County, OH (Battelle, 2011). A test well was drilled at the site, reaching a total depth of 2555 m below ground surface (bgs) in February 2007. A preliminary assessment of the geology at the site identified three deep carbonate and sandstone formations between 1800 and 2500 m below the surface for CO_2 injection. Wireline logs indicated the three target injection zones had porosities between 2%–10%. Analysis of cores from these zones confirmed the rock had low porosity (below 5%) and very low permeability (below 9.9×10^{-16} m^2).

A series of injection tests were completed in each of the three target formations. Hydraulic analysis predicted injection rates of nearly 50 t per day at injection pressures below 13.8 MPa; however, actual flow rates observed were less than 20 t per day at more than 25.6 MPa. Successive hydrochloric acid “stimulation” treatments were attempted to clean the injection zones, yet the efforts failed to improve injectivity. The injection tests at the Burger well revealed that the three targeted geologic formations (at this location) did not have sufficient porosity and permeability for completing the small-scale injection of 3000 t of CO_2, and the well was plugged and abandoned in 2010. This case highlights the operational issue that some CO_2 point source emitters may experience, sub-optimal injectivity/capacity conditions below their plant.

The State of Ohio had regulatory oversight over injection wells in the state, and a well stimulation operation was not completed due to Ohio EPA Underground Injection Control (UIC) permit restrictions. The three formations tested by the MRCSP are routinely stimulated (fractured) for oil and gas production in the area, a step that may have led to a successful CO_2 injection campaign at the site.

1.1.2. FutureGen 2.0

The FutureGen 2.0 Project planned to capture approximately 22 million metric tonnes (Mt) of CO_2 emissions over 20 years (1.1 Mt/year) from a coal-fired power plant in Meredosia, Illinois. A 30-mile long CO_2 pipeline was to be built to transport the captured CO_2 to a permanent storage site in Morgan County, Illinois, where it was to be injected 1240 m underground into the Mt. Simon Sandstone through four horizontal injection wells. A stratigraphic borehole was drilled at the storage site in 2011, and after extensive characterization, the project team determined that the proposed storage reservoir was several hundred feet thinner than anticipated (Kelly et al., 2012). Additionally, the permeability of the Mt. Simon Sandstone was found to be as much as an order of magnitude lower than observed in other wells throughout the Illinois basin (Sullivan et al., 2015).

Numerical modeling simulations determined that the optimum injection design would use four horizontal injection wells (with 460 m and 760 m long horizontal sections) to limit pressure increases within the reservoir. This resulted in a complex injection well design that required four separate underground injection control permit applications (one for each injection well) (Gilmore et al., 2016). Additionally, the thinner-than-expected storage reservoir necessitated acquiring additional pore space rights to accommodate the expanded volume of the planned CO_2 plume. Collectively, these undesirable attributes were resolved by the project team; however, the cost and complexity of the effort was not conducive to widespread deployment of CCUS.

1.1.3. Wallula

The world’s first injection of supercritical CO_2 into a continental flood basalt was performed by the U.S. Department of Energy’s Big Sky Regional Carbon Sequestration Partnership (BSRCSP) in Washington State in 2013. The CO_2 injection well was drilled to 1253 m bgs in early 2009 (McGrail et al., 2011). Subsequent hydrogeologic testing identified decreasing permeability with depth. A total of 10 test intervals were evaluated, and a candidate CO_2 injection interval was identified between the depth interval of ~828 and 875 m bgs. This interval contained three brecciated interflow zones that had moderate permeability (1.5×10^{-15} to 7.4×10^{-14} m^2) and were bounded by thick flow interiors that had low (10^{-18} m^2) permeability.

A 25-day CO_2 injection campaign was initiated across all three zones on July 17, 2013 and was terminated just prior to reaching the Class V injection permit limit of 1000 t (McGrail et al., 2014). Injection rates average 40 t per day and were generally uniform throughout the injection period. Daily injection rates were intentionally limited to keep the downhole test interval pressure within 2.8 MPa above hydrostatic reservoir conditions. Within the first 24 h of injection, the downhole pressure increased rapidly by 2.1 MPa. The pressure buildup leveled off briefly before continuing to increase steadily at a lower rate of approximately 0.028 MPa per day over the remainder of the injection period. The challenge associated with the BSRCSP project was that, although the basalt reservoir had good porosity 15 %–25 %, the permeability was such that it took over three weeks to inject approximately 1000 t of CO_2, or approximately 40 % of the daily emissions from a 1000 MW h coal-fired power plant. Despite the benefits of injecting CO_2 into basalt (e.g., mineral trapping of CO_2 via calcite precipitation), the

low permeability observed at the BSRCSP site could not support commercial-scale injection rates without improved injectivity within the storage reservoir.

1.2. Previous research evaluating the use of hydraulic fracturing in saline CO_2 storage reservoirs to increase injectivity and capacity

Ghaderi et al. (2009) conducted simulations to assess the impact of hydraulic fracturing on CO_2 injectivity for a case of injection of 1 Gt of CO_2 into the Nisku aquifer in the Wabamun Lake Area of Alberta, Canada, over a period of 50 years. In the simulation, the aquifer was homogenous, isotropic, and isothermal, with a thickness of 100 m, a permeability of 1.0×10^{-13} m^2, and 12 % porosity. They performed simulations for four cases: a vertical well, a horizontal well, and vertical and horizontal wells with hydraulic fractures. For the vertical well, a hydraulic fracture of 400 m (half-length) was simulated. For the horizontal well case (length of 3000 m), four staggered 100 m (half-length) fractures were used in the simulation. A porosity of 15 % and permeability of 1.5×10^{-12} m^2 were assigned to thin grid blocks used to model the fractures. The simulation results indicated that a horizontal well would increase the CO_2 storage capacity by about 63 % over that of the vertical well. The horizontal well with fractures would increase the storage capacity by approximately a factor of two relative to the non-fractured vertical well and by about 35 % compared to the non-fractured horizontal well. These simulation results suggest that the use of hydraulic fracturing in combination with horizontal wells could result in a significant increase in CO_2 storage capacity.

Using an equation-of-state compositional simulator, CMG-GEM (Nghiem et al., 2004), Raziperichikolaee et al. (2013) performed simulations to evaluate the effect of hydraulic fracturing on the long-term storage of CO_2 in the Rose Run sandstone aquifer in the Ohio River Valley. Based on their results, it was shown that hydraulic fracturing could significantly enhance injectivity and increase the practical storage capacity of thin, low permeability, saline aquifers. They also determined that the enhanced permeability in the aquifer around the wellbore significantly reduces the long-term trapping of CO_2 for a normal faulting stress regime. By limiting the fracture height, post-injection immobilization of CO_2 could be enhanced compared to an aquifer that was fractured throughout its full height.

1.3. New simulations to evaluate the use of hydraulic fracturing to improve the performance of a class VI commercial-scale CO_2 storage reservoir

We used the FutureGen 2.0 site for the case study because, as previously mentioned, the project was granted four Class VI UIC permits and significant characterization data are available to support supplemental analyses (FutureGen, 2015). The project is also an example of how a site's CO_2 injectivity may be reduced from the initial estimate to a less-than-economic value once site-specific geologic and petrophysical properties becomes known after drilling a stratigraphic well. While the research team was ultimately successful in developing a plan to overcome limited injectivity by deploying four lateral injection wells, this solution would lead to a significant capital cost increase for the project through the drilling and completing of additional wells, all of which would undergo an additional regulatory burden, as each well would require its own EPA UIC Class VI permit. Hence, in this case study we wish to study how hydraulic fracturing could have been used as an alternative solution to achieve the original operational objectives.

For our analysis, we retained the original goal of the FutureGen 2.0 project to inject 22 Mt over 20 years, with an ideal injection rate of 1.1 Mt/yr and maximum injection rate of 2.0 Mt/yr at a maximum injection pressure of 15.48 MPa.

1.3.1. Methods

The model domain and geologic properties are based off of previous work from the FutureGen 2.0 project (FutureGen, 2015; White et al., 2016). The model consists of a layered and stacked system from the bottom of the storage interval through the top of the primary seal. The model uses a local cartesian coordinate system and has dimensions of 54,864 m in the x-direction, 43,536.41 m in the y-direction, and 254.81 m in the z-direction. The elevation range for the storage interval (in the z-direction) is -903.12 m to -1,157.94 m. The coordinate system is aligned so that the x-direction lies in the same direction as the least principle stress as identified by the FutureGen 2.0 site characterization efforts (FutureGen, 2015). The z-direction is down and the y-direction is orthogonal to the other two directions.

Following the work of the FutureGen 2.0 project, our storage system lies within the Illinois basin and targets the thick sedimentary units overlying the Precambian basement rock. In our model, we define a storage or injection zone and an overlying primary seal. The storage interval consists primarily of the thick Mt. Simon sandstone but also includes the Elmhurst sandstone member of the Eau Claire. The overlying primary seal is the Lombard dolomite member of the Eau Claire. We first define the model domain and how it can be altered to present different scenarios, then we describe the simulation matrix that comprises the different scenarios.

The reservoir model consisted of 26 total layers (in the z-direction) with four in the Lombard, seven in the Elmhurst, and fifteen in the Mt. Simon. Each layer has a distinct thickness, porosity, and permeability (Fig. 1). The unit thickness, porosity, permeability, and other parameters were taken from the FutureGen 2.0 Final Report and the corresponding EPA UIC Class VI permit document (FutureGen, 2015; EPA, 2013). Note that some layers were combined, specifically those within the Lombard, to reduce simulation time.

Geomechanical analysis was conducted at the FutureGen 2.0 site for general characterization and as part of the UIC class VI permit (EPA, 2013; FutureGen, 2015; White et al., 2016). The analysis looked at well-based and sample-based analysis of stress indicators and rock properties. They found that the direction of maximum horizontal stress (S_H) was N79.9 °E and that S_H was larger than the least principle stress (S_h). While the third principle stress was the vertical stress (S_v), they could not determine if it was greater or less than S_H. Although recent work has shed more insights into the nature of the tectonic regime at the site (Appriou, 2019), the faulting style is treated as an unknown and something to account for in this study. The regime is important for hydraulic fracture development because, while the fracture opens due to failure in the direction of the least principle stress, the fracture would dominantly grow in the direction of the greatest principle stress. Two scenarios were developed where the faulting regime was varied. One scenario has a normal faulting regime ($S_v > S_H > S_h$) and the other has a strike-slip regime ($S_H > S_v > S_h$). Given that the relative magnitude between S_H and S_v is unknown, we looked at several ratios of fracture area (i.e., the fracture height multiplied by its width) to characterize the impact of fracture geometry on the results. While a detailed model could be used to predict the fracture properties (e.g., Adachi et al., 2007), given the significiant uncertainties in parameters and that the goal of this work is to characterize injectivity and capacity, we used a simplified approach.

To determine the initial and maximum injection pressure we use observations from the initial FutureGen 2.0 site characterization and Class VI permit application (FutureGen, 2013, 2015). Fig. 2 shows the subsurface pressures for our site. Plotted for reference are the overburden gradient (solid black line) at 2.26×10^{-2} MPa/m (1.0 psi/ft) and fresh water gradient (dashed black line) at 9.80×10^{-3} MPa/m (0.433 psi/ft). During drilling of the stratigraphic well fluid pressure was measured at eight points (green asterisks). These measured pressures have a corresponding pressure gradient of 1.02×10^{-2} MPa/m (0.45 psi/ft). Thus, the initial condition for reservoir pressure at the location of the horizontal wells (−1049 m) is 12.34 MPa (black circle on Fig. 2). To calculate the maximum injection pressure we use the analysis and reccomendation from the Class VI permit and use a fracture gradient of 1.47×10^{-2} MPa/m (0.65 psi/ft) with an additional

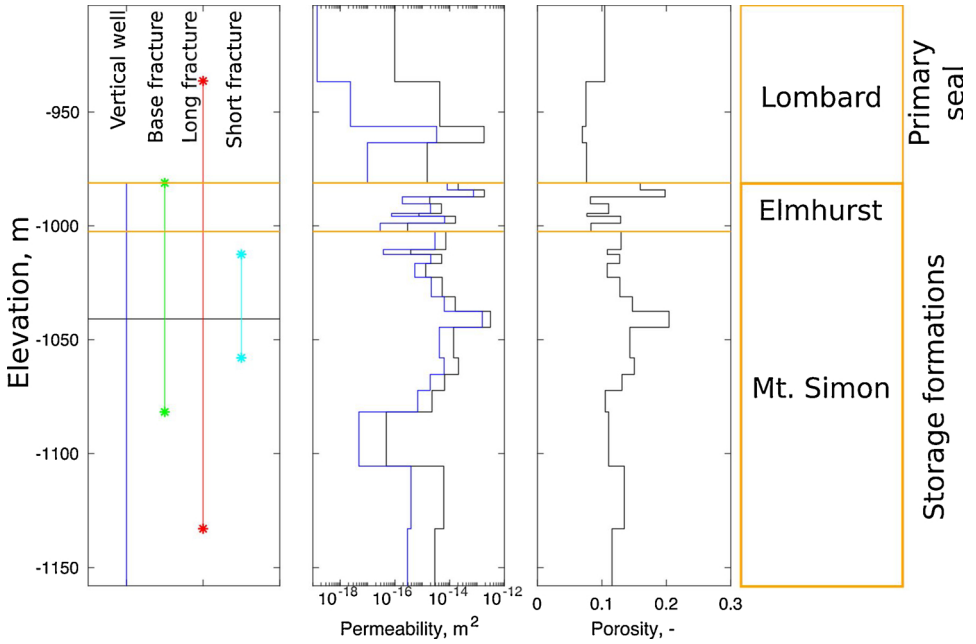


Fig. 1. Unit names, petrophysical properties, well positions, and fracture intervals are shown on this figure. (left plot) The position of the vertical well (blue line) and horizontal well (black line) are shown as a function of depth. The hydraulic fracture heights studied in the simulation are the base case (green line), the cases of area increase by fracturing through most of the injection interval and into the bottom of the primary seal (red line), and the case of an underperforming fracture (cyan). (middle plot) Permeability in the x- and y-direction (black line) and in the z-direction (blue line) is plotted as a function of elevation. (right plot) Porosity is plotted as a function of depth. (right illustration) shows the key stratigraphic units.

engineering factor of 0.9. Because we allow for fracturing into the Elmhurst we define the maximum injection pressure as the value at the base of the Lombard (-981.15 m) and calculate the maximum injection pressure to be 15.48 MPa.

To perform the study, simulations were conducted using STOMP-CO2 (Subsurface Transport Over Multiple Phases), a multi-phase subsurface flow and transport simulator developed at Pacific Northwest National Laboratory. This code has been extensively developed and applied to simulate fate and transport of CO_2 and brine in CCUS systems and has been verified against other codes used for CCUS in the GeoSeq code comparison study (Pruess et al., 2004; White et al., 2016; Yonkofski et al., 2019). STOMP is also stated as an option for use in the modeling component required as part of the application to obtain a permit to inject CO_2 into the subsurface for CCUS (EPA, 2013a, b). For details on the model's theory and development see White et al. (2012). The reservoir model was adapted from White et al. (2016), which contains most of the model assumptions and parameters, with the exception for the changes described below.

The following considerations motivated the grid generation: represent the three geologic units, encompassed a large enough area to avoid boundary effects, provide a consistent grid pattern to be used by all simulations, and provide fine enough detail at the fracture grids to simulate hydraulic fracturing. To accommodate the latter requirement, variable grid sizes were used in the x- and y-direction. To accommodate the variable thickness of the geologic units (and layers within), variable grid sizes were also used in the z-direction. The total number of grid cells was 340,808 (x-dir: 226, y-dir: 58, z-dir: 26). The smallest grid size in the y-direction is 15.24 m to accommodate the finer resolution needed to simulate the fractures, and increases to 8,090.00 m at the boundaries. In the x-direction, sizes ranged from 0.317 m at the cells containing the hydraulic fractures to 2286 m at the boundaries. The cell size follows the layer size in the z-direction and is between 1.22 m and

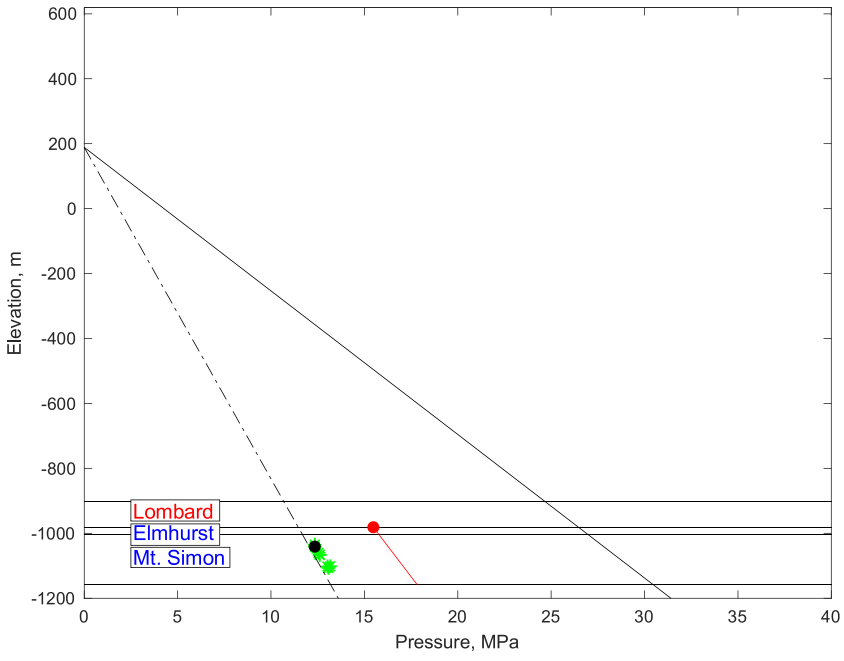


Fig. 2. Plot of the fluid, rock, and fracture pressures as a function of depth. The pressures corresponding to a 2.26×10^{-2} MPa/m (1.0 psi/ft) overburden gradient (solid black line) and a 9.80×10^{-3} MPa/m (0.433 psi/ft) freshwater gradient (dashed black line) are plotted for reference. Fluid pressure in the Mt. Simon (green asterisks) was measured at eight points in the FutureGen 2.0 stratigraphic well (FGA-1). A fracture gradient of 0.65 psi/ft was used in this model (red line). The fracture pressure at the bottom of the Lombard (15.48 MPa) is the maximum injection pressure used in our simulations.

Table 1
Experiment matrix and key well and fracture properties.

Experiment name*	Well orientation	Injection length, m	Number of fractures	Fracture dimension (x,y,z), m	Fracture permeability (x,y,z), m^2	Fracture porosity, -
Case-v	Vertical	176.78	0	NA	NA	NA
Case-h-3k	Horizontal	914.4	0	NA	NA	NA
Case-h-6k	Horizontal	1,828.8	0	NA	NA	
Case-h-3k-4f-b	Horizontal	914.4	4	0.317, 91.44, 100.58	3.06×10^{-13}, 2.63×10^{-10}, 2.63×10^{-10}	0.2067
Case-h-3k-4f-ss	Horizontal	914.4	4	0.317, 60.96, 100.58	3.06×10^{-13}, 2.63×10^{-10}, 2.63×10^{-10}	0.2067
Case-h-3k-4f-nf	Horizontal	914.4	4	0.317, 213.36, 100.58	3.06×10^{-13}, 2.63×10^{-10}, 2.63×10^{-10}	0.2067
Case-h-6k-8f	Horizontal	1,828.8	8	0.317, 91.44, 100.58	3.06×10^{-13}, 2.63×10^{-10}, 2.63×10^{-10}	0.2067
Case-h-3k-4f-ai	Horizontal	914.4	4	0.317, 91.44, 196.60	3.06×10^{-13}, 2.63×10^{-10}, 2.63×10^{-10}	0.2067
Case-h-3k-4f-ad	Horizontal	914.4	4	0.317, 91.44, 45.42	3.06×10^{-13}, 2.63×10^{-10}, 2.63×10^{-10}	0.2067
Case-h-3k-4f-pi	Horizontal	914.4	4	0.317, 91.44, 100.58	6.12×10^{-13}, 5.26×10^{-10}, 5.26×10^{-10}	0.2073
Case-h-3k-4f-pd	Horizontal	914.4	4	0.317, 91.44, 100.58	3.06×10^{-13}, 3.32×10^{-11}, 3.32×10^{-11}	0.2054
Case-h-6k-8f-ai-pi	Horizontal	1,828.8	8	0.317, 91.44, 196.60	6.12×10^{-13}, 5.26×10^{-10}, 5.26×10^{-10}	0.2073

* The naming convention for the Experiment names is: (v) vertical well, (h) horizontal well, (3k) 3000 foot lateral (914.4 m) (6k) 6000 foot lateral (1,828.8 m), (4f) four fractures, (8f) eight fractures, (b) base case, (ss) strike-slip regime, (nf) normal fault regime, (ai) area increase, (ad) area decrease, (pi) permeability increase, (pd) permeability decrease.

19.66 m.

The improvement in hydraulic conductivity or equivalent permeability of the reservoir after hydraulic fracturing is unknown and depends on the lithology, stress state, rock properties, and ultimately on the design and success of the hydraulic fracture operation. Because the goal of this work is to characterize the impact that hydraulic fracturing could have on geologic carbon storage operational goals, simple relationships are used to estimate equivalent porosity and permeability for the grid cells that contain the fracture. For a given scenario with factures we define specific grids as belonging to the "fracture" and then assign them properties representative of a fracture. The length of any cell containing a fracture is: x-direction: 0.317 m, y-direction: 15.24 m, and the z-direction is set depending upon which layers the particular fracture is defined over. The z-direction cell size is between 3.048 m and 33.528 m. The total number of x- and y-direction grids that are assigned to be a fracture depends on the scenario. This will allow us to define different fracture geometries based on the tectonic stress-state being normal faulting (nf) or strike-slip faulting (ss) or based on uncertainty in fracture area deriving from the hydraulic fracture operation, which could yield an area increase (ai) or area decrease (ad) from some base case (b).

For those grids assigned as a fracture, we next define their permeability (k) values and their porosity. Recall that the least principle stress is in the x-direction and thus, flow in this direction will be governed by flow in series and will be dominated by matrix flow. For flow in the x-direction, we define the permeability as $k_x = 3.06 \times 10^{-13}\,m^2$ for the base case. This value is equal to the x-direction permeability of the layer in the Mt. Simon that contains the horizontal well. To approximate the fluid flow occurring in the fracture and porous media surrounding the fracture contained within the cell we assumed flow in parallel in the y- and z-directions and must calculate the effective permeability for the fracture (k_{eff}). We use $k_y = 3.06 \times 10^{-13}\,m^2$, and $k_z = 1.55 \times 10^{-13}\,m^2$ as the permeability for the porous media. Given the uncertainties in rock properties, specifics of the hydraulic fracturing operation, large span in the scale of simulated phenomena, and limitations on computational resources we employ simple constitutive models to relate the hydraulic fracture properties to changes in rock permeability. To set the base case we assume flow through a fracture with aperture (B) of 1000 μm. Equating Reynolds equation with the Darcy equation (Snow, 1965; Bear, 1972; Zhang et al., 2018) we have:

$$k_f = \frac{B^2}{12}$$

Thus, the equivalent permeability of the fracture is $k_f = 8.33 \times 10^{-8}\,m^2$. However, the area of the fracture is comparatively small to the area of the grid that contains both rock and the fracture, so we account for that by calculating an effective permeability using the following equation for flow in multiple parallel units (Snow, 1965; Bear, 1972; Zhang et al., 2018):

$$k_{eff} = \frac{k_1 A_1 + k_f A_f + k_2 A_2}{A_{tot}}$$

This equation normalizes the permeability from the fracture (f) and the two sides of permeable rock bounding the fracture (1 and 2) by their relative areas, where A_{tot} is the sum of areas 1, 2, and f. The resulting k_{eff} is $2.63 \times 10^{-10}\,m^2$. To account for uncertainty and study the impact of over or underestimating the change in permeability of the reservoir due to hydraulic fracturing, we define a permeability increase (pi) scenario with $k_{xpi} = 6.12 \times 10^{-13}\,m^2$ and $k_{effpi} = 5.26 \times 10^{-10}\,m^2$, and a permeability decrease (pd) scenario with $k_{xpd} = 3.06 \times 10^{-13}\,m^2$ and $k_{fpd} = 3.32 \times 10^{-11}\,m^2$. The pi case would have a fracture aperture equivalent to 1260 μm and the pd case would have a fracture aperture equivalent to 500 μm. To calculate the effective porosity for cells containing the fracture, the fracture volume was taken into account along with the rock porosity.

The orientation of wells will be either vertical (one case) or

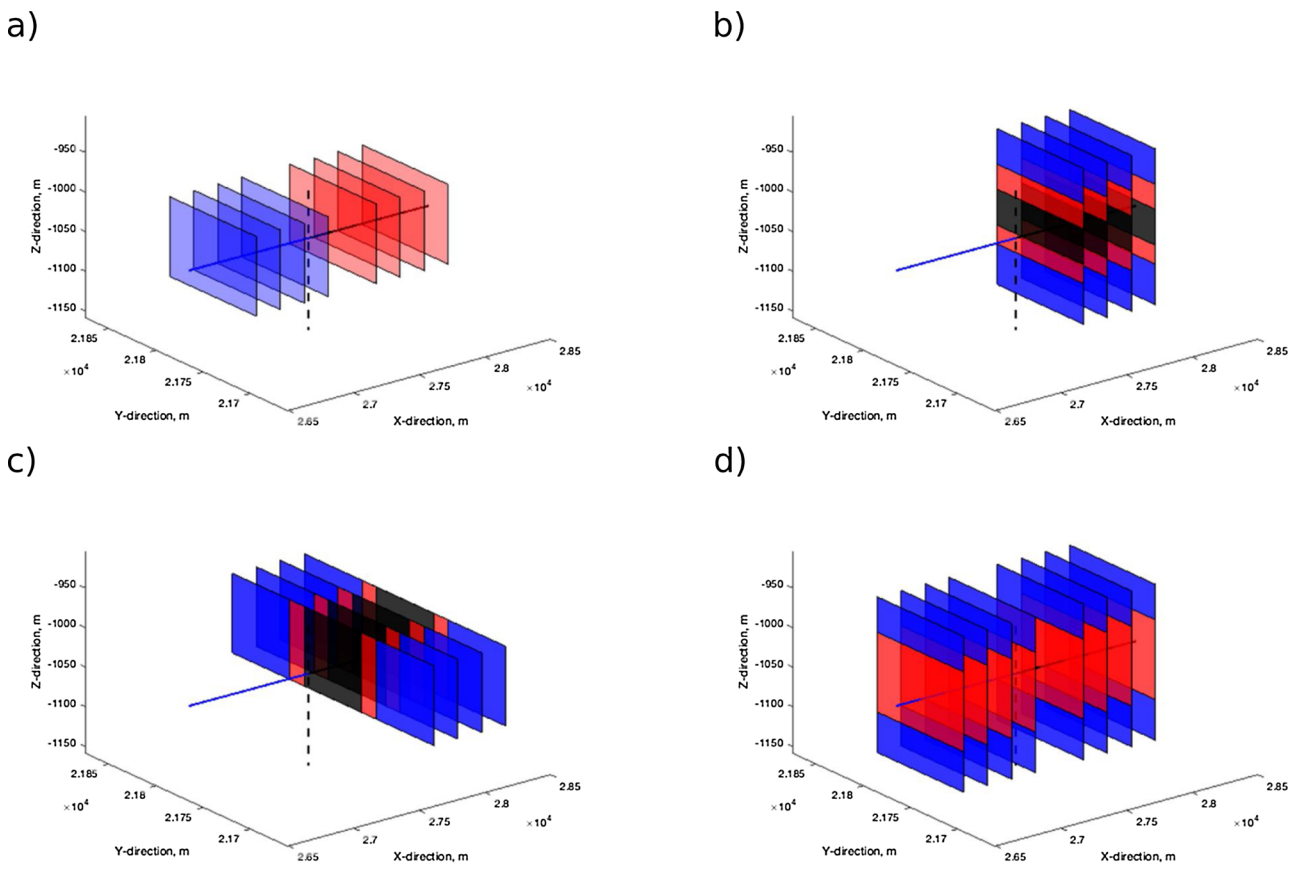


Fig. 3. Well and fracture geometries for the various cases studied in this work. a) Case-v: vertical dashed black line. Case-3k: horizontal black line. Case-3k-4f-b: horizontal black line and four red rectangles. Case-6k: horizontal black line and horizontal blue line. Case-6k-8f: horizontal black line, four red rectangles, horizontal blue line, and four blue rectangles. b) Vertical black dashed line and horizontal blue line shown for reference. All three of the following cases use the horizontal black line and rectangles are inclusive of the area from the well to the edge of their respective colors. Case-3k-4f-b: four red rectangles. Case-3k-4f-ai: four blue rectangles, Case-3k-ad: four black rectangles. c) Vertical black dashed line and horizontal blue line shown for reference. All three of the following cases use the horizontal black line and rectangles are inclusive of the area from the well to the edge of their respective colors. Case-3k-4f-b: four red rectangles. Case-3k-4f-nf: four blue rectangles, Case-3k-ss: four black rectangles. d) Vertical black dashed for reference. All three of the following cases use the horizontal black line and horizontal blue line and rectangles are inclusive of the area from the well to the edge of their respective colors. Case-6k-8f: eight red rectangles. Case-6k-8f-ai-pi: eight blue rectangles. Note: Case-3k-4f-pi and Case-3k-4f-pd share the same area as Case-3k-4f-b.

Table 2
Results for the cases studied.

Experiment name*	Total CO_2 injected, Mt	Injection rate at end of injection interval, Mt/yr	Time to injection rate below goal rate (1.1 Mt/yr), yr
Case-v	13.14	0.68	1.1×10^{-4}
Case-h-3k	14.81	0.74	5.8×10^{-2}
Case-h-6k	17.54	0.85	6.4×10^{-1}
Case-h-3k-4f-b	16.25	0.81	1.8×10^{-1}
Case-h-3k-4f-ss	16.18	0.81	1.7×10^{-1}
Case-h-3k-4f-nf	16.56	0.82	2.4×10^{-1}
Case-h-6k-8f	19.10	0.91	1.5
Case-h-3k-4f-ai	19.56	0.96	9.0×10^{-1}
Case-h-3k-4f-ad	14.84	0.74	5.7×10^{-2}
Case-h-3k-4f-pi	16.25	0.81	1.8×10^{-1}
Case-h-3k-4f-pd	16.19	0.81	1.7×10^{-1}
Case-h-6k-8f-ai-pi	22.44	1.06	6.9

* The naming convention for the Experiment names is: (v) vertical well, (h) horizontal well, (3k) 3000 foot lateral (914.4 m) (6k) 6000 foot lateral (1,828.8 m), (4f) four fractures, (8f) eight fractures, (b) base case, (ss) strike-slip regime, (nf) normal fault regime, (ai) area increase, (ad) area decrease, (pi) permeability increase, (pd) permeability decrease.

horizontal (remaining case). Fig. 1 (left plot) shows the length of the vertical well (blue line), location of the horizontal well (black line), and the height of the three different fractures studied in different scenarios (green, red, and cyan). The length of the injection interval for the vertical well is over the entire Elmhurst and Mt. Simon intervals (176.78 m). The location of the horizontal injection well is within the high permeability streak in the Mt. Simon. The length of the horizontal wells will either be 914.4 m (3000 ft) or 1,828.8 m (6000 ft) and the case name will be appended with a −3k or −6k respectively. For each respective length there will be four fractures of uniform dimension (for a given case). Thus, for -3k simulations that have been hydraulically fractured there will be four fractures and for the -6k simulations there will be eight fractures. The height of the three possible hydraulic fractures is 100.58 m for the base, 45.42 m for the shortest, and 196.60 m for the tallest. Table 1 and Fig. 3 show how the well and fracture orientation changes for each of the cases.

Table 1 summarizes the key parameters defining each of the cases simulated in the study. In the results section we will compare the injectivity and capacity for the vertical well, horizontal wells, and horizontal wells with fracture. We will determine how tectonic regime, which may alter the fracture's geometry, affects the results. We will also look at how uncertainty in the fracture area and in the fracture permeability may affect the results.

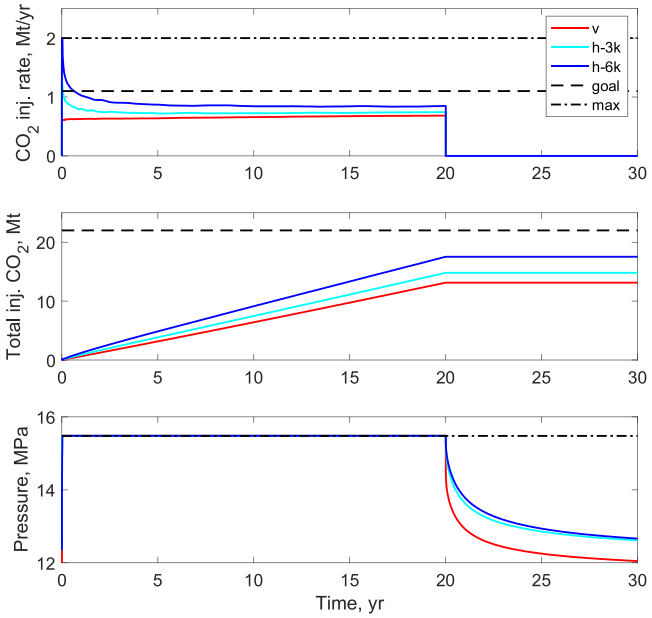


Fig. 4. Plot showing CO_2 injection rate, total CO_2 injected, and pressure at the injection well for Cases consisting of vertical or horizontal wells of different length without fractures (Table 2). The simulations are compared against the maximum allowable (rate and pressure) and optimum (rate and total amount) value.

2. Results

Table 2 presents the results for the twelve cases studied in this work.

2.1. Vertical and horizontal wells

This section compares the cases with different well orientations and lengths, and no hydraulic fractures (case-v, case-h-3k, and case-h-6k). The behavior for the three scenarios follows the same general trends (Fig. 4). Injection begins at the maximum injection rate (2.0 Mt/yr) but as the injection pressure reached the maximum allowed pressure, the injection rates were reduced. Injection continued until 20 years had elapsed and none of the cases were able to achieve the injection goal of 22 Mt (Table 2). The vertical well performed the worst of all cases, achieving only 60 % of the injection goal, the 3k horizontal well showed a slight increase in performance at 67 %, and the 6k well was significantly better at 80 % of the goal of 22 Mt.

After injection stops at 20 years, the pressure at the well decreased for the next decade. The vertical well had a significantly faster initial pressure decrease when compared to the two horizontal well cases (which were essentially identical). The long-term rate of pressure decrease over time for the three cases is similar and all remain above the initial pressure at the end of the simulations (Fig. 4).

2.2. Unfractured and fractured horizontal wells

This section compares the cases of horizontal wells of different lengths, with and without hydraulic fractures (case-h-3k, case-h-3k-4f-b, case-6k, case-6k-8f), and the most optimistic case (case-6k-8f-ai-pi). Fig. 5 shows the evolution of the CO_2 plume during the injection period for case-h-3k-4f-b. The behavior for the horizontal cases follows the same general trends observed in the previous section and with each other (Fig. 6). For all cases, injection begins at the maximum injection rate (2.0 Mt/yr), but as the injection pressure reached the maximum allowed pressure, the injection rates were reduced. In comparing the cases, both longer well length and the presence of hydraulic fractures had a positive impact in allowing for a longer time until injection rates fell below the goal rate with higher final injection rates (Table 2). For example, the case-3k CO_2 injection rate fell below the goal rate after 21 days of injection (5.8×10^{-2} yr), but the longer case-6k rate did not fall below the rate until 234 days (6.4 × 10-1 yr). This trend is also observed in the best case scenario (case-h-6k-8f-ai-pi) where, in addition to having fractures, the area and permeability have been increased.

While there was an improvement in injectivity across all scenarios over the base case of a vertical injection well, only the best-case scenario was able to achieve the total injection amount goal. The next best was case-6k-8f, which allowed for 87 % of the goal amount.

The pressure decline post-injection at the well was nearly identical for all cases studied in this section and had the same characteristic initially-sharp decrease in pressure for the first three years, after which the rate of pressure decay flattens out but is above (but near) the initial pressure at the end of the simulations (Fig. 6).

2.3. Uncertainty in stress state

Uncertainty in the underlying stress state at the FutureGen 2.0 site motivated comparing the three cases with different fracture geometries (case-h-3k-4f-b, case-h-3k-4f-nf, and case-h-3k-4f-ss). The operational limitation was to assume hydraulic fracturing would yield a vertical fracture of the same height for all three cases (100.58 m) to avoid fracturing into the seal, but depending on the stress state, the fracture would have different lateral growth lengths. The total fracture width would be 91.44 m for the base case, 60.96 m for the strike-slip case, and 213.66 m for the normal faulting case (Fig. 3c). These dimensions yield fracture areas for a single fracture of 9192 m^2 for the base case, 6131 m^2 for the strike-slip case, and 21,460 m^2 for the normal fault case. The results show no significant difference in injection rate, total amount injected, or pressure at the well over time (Fig. 7). The injection pressure limitation is the controlling factor on injection rate and adjusting the fracture geometry does not change the relationship between how the well intersects the fracture.

2.4. Uncertainty in fracture quality

There is significant uncertainty with how a hydraulic fracturing job will proceed and what the resulting system's properties will be. To capture some aspects of that uncertainty we presume the fracture area may be perturbed (specifically by varying the height) and the effective fracture permeability may be altered. Fig. 8 shows the results of these alterations to the base case (case-h-3k-4f-b). The CO_2 injection rate shows that only the changing of the fracture area has an impact to the simulations. The fracture heights were chosen to be two times the base for the area increase and half the base for the area decrease but there is not a correspondingly linear change in total CO_2 injected. The area increase has a much larger impact (3.32 Mt more injected than the base case) than the area decrease (1.41 Mt less injected than the base case). The well pressure over time for the three cases is essentially identical.

3. Discussion

Over the last decade, significant investments have been made domestically and internationally to develop carbon capture, utilization, and storage technologies for their potential to mitigate environmental impacts associated with climate change due to atmospheric release of CO_2 from anthropogenic sources. Deep saline aquifers remain the primary storage target, as they represent as much as 98 % of the onshore storage capacity within the US. Despite years of investments that have included both pilot- and commercial-scale storage projects, there are several examples of CO_2 storage projects that were negatively impacted by poor injectivity and/or capacity with their proposed storage reservoirs. Our work addresses three critical questions that help answer if hydraulic fracturing can be used to safely enhance the performance (injectivity and storage capacity) of CO_2 storage reservoirs. Those

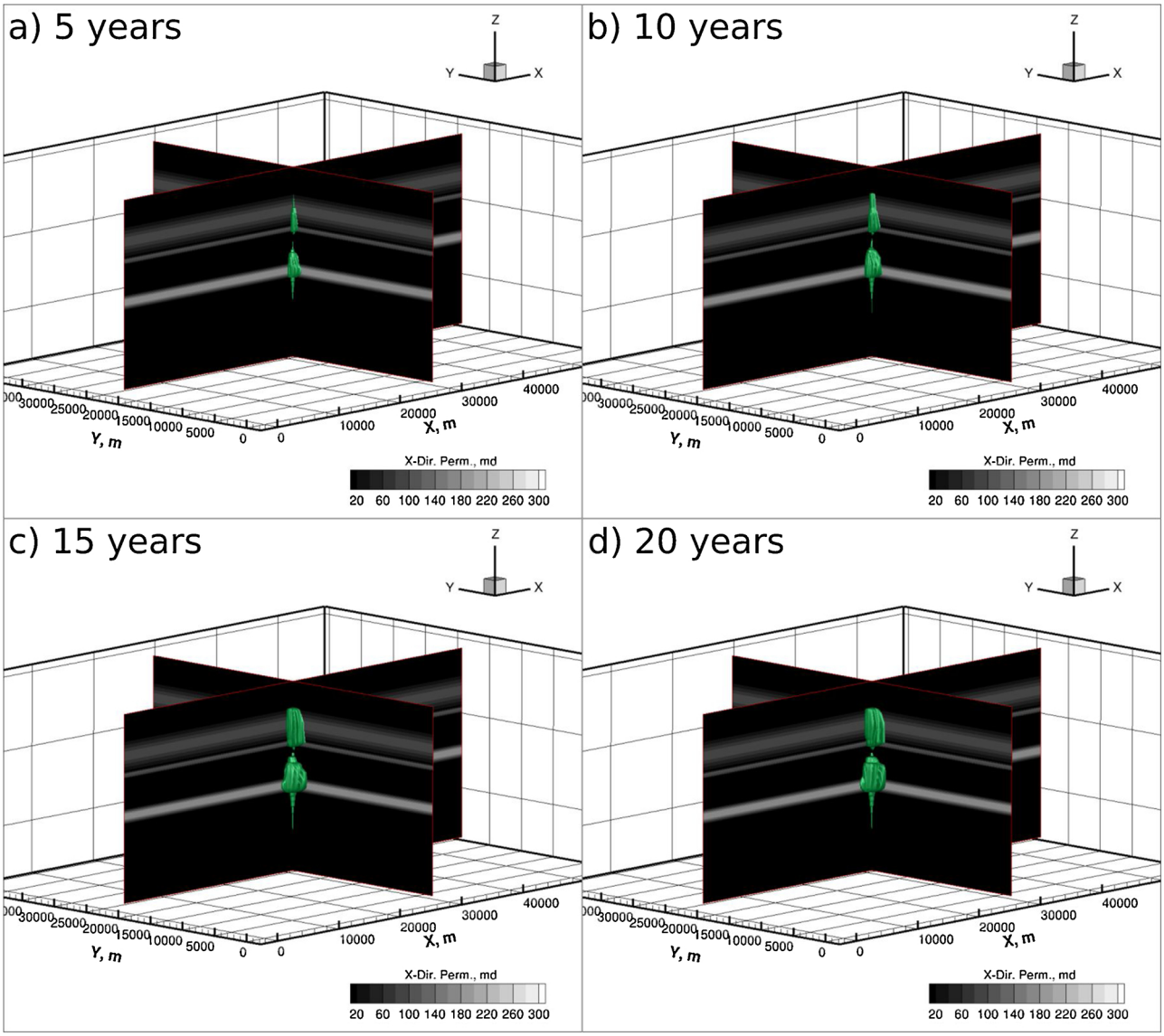


Fig. 5. Case-h-3k-4f-b. Plots showing an isosurface (green) of CO_2 saturation equal to 0.01. Slices in the x-direction and y-direction show the x-direction permeability. During injection, the CO_2 plume expands in the high permeability zones intersected by the fractures.

questions are:

- Did the hydraulic fracturing operation increase the long-term risk of the project?
- Did we achieve the operational objective?
- What is the benefit of the enhanced injectivity/capacity when compared to the increased complexity of the well?

3.1. Risk

While the hydraulic fracturing operation itself will have inherent risks (e.g., possibly inducing seismicity or fracturing through the caprock) these are not studied in our present work and we only briefly discuss them so that we can focus on the long-term risks as compared to the baseline (vertical) well case. The long-term project risks are how the CO_2 footprint (or plume size) changes as a function of the hydraulic fracturing and what is the disposition of the CO_2 over time when compared to the base case.

Some of the primary risks associated with hydraulic fracturing operation are fractures intersecting abandoned or converted oil and gas wells, or fractures "running away" and fracturing the caprock. Given the ubiquitous use of hydraulic fracturing during the development of unconventional oil and gas wells, hydrocarbon production wells serve as excellent analogs for studying potential environmental impacts of hydraulic fracturing. Analysis of 315,000 oil and gas wells completed in Texas and Ohio uncovered only 396 groundwater contamination incidents, none of which were related to hydraulic fracturing (Kell, 2011). Therefore, when performed properly, the risk associated with fracture interactions with abandoned and converted oil and gas wells has been demonstrated to be very low.

Another potential risk is that pressure increases due to the large volumes of CO_2 injected into underground storage reservoirs could create conditions that promote new fractures or reopen existing fractures in the caprock seal. To evaluate this, Pan et al. (2013) conducted modeling studies using multiphase flow models coupled with geomechanical models, and made the following important conclusions: 1) caprock sealing integrity could be maintained by limiting the injection pressure to the minimal principal stress, 2) effective methods for early detection of fracture propagation through a caprock layer include pressure and deformation or stress field monitoring, and 3) hydraulically-induced fractures induced by CO_2 injection are more likely to propagate horizontally than vertically and remain contained within

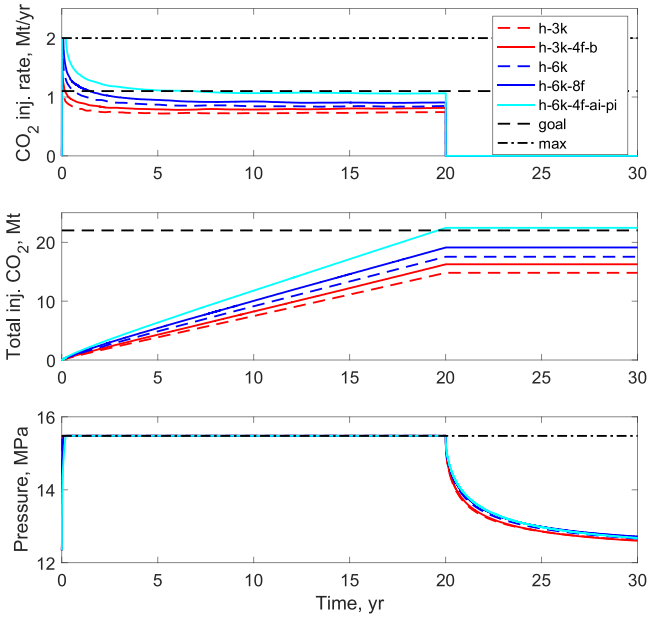


Fig. 6. Plot showing CO_2 injection rate, total CO_2 injected, and pressure at the injection well for Cases consisting of horizontal wells of different length with and without fractures (Table 2). The simulations are compared against the maximum allowable (rate and pressure) and optimum (rate and total amount) value.

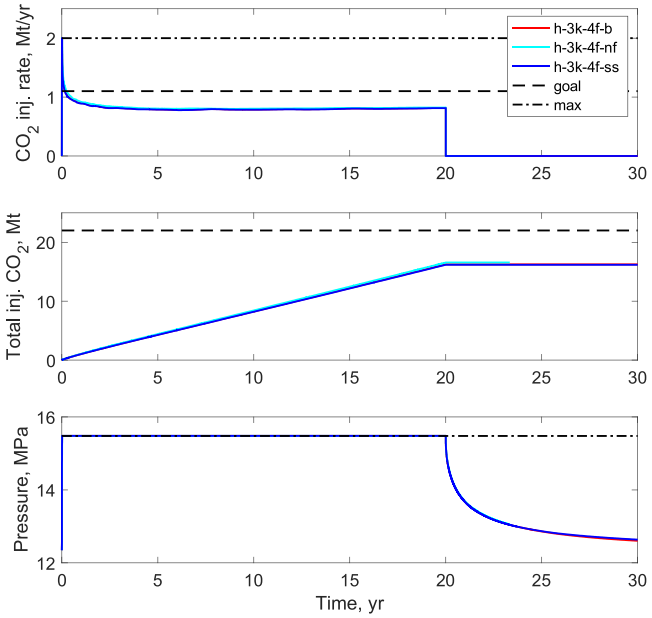


Fig. 7. Plot showing CO_2 injection rate, total CO_2 injected, and pressure at the injection well for fractured horizontal wells with different fracture geometries related to possible stress states at the FutureGen 2.0 site. The simulations are compared against the maximum allowable (rate and pressure) and optimum (rate and total amount) value. Note that the normal fault (nf) scenario was terminated early to conserve computational resources but has a nearly identical trends to the other cases.

the injected storage zone. Additionally, this horizontal fracture propagation will likely have positive effects on injectivity and storage capacity of the formation.

3.1.1. Footprint

The CO_2 plume footprint is important as it represents the area where buoyant CO_2 remains and could act as a source of upward migrating

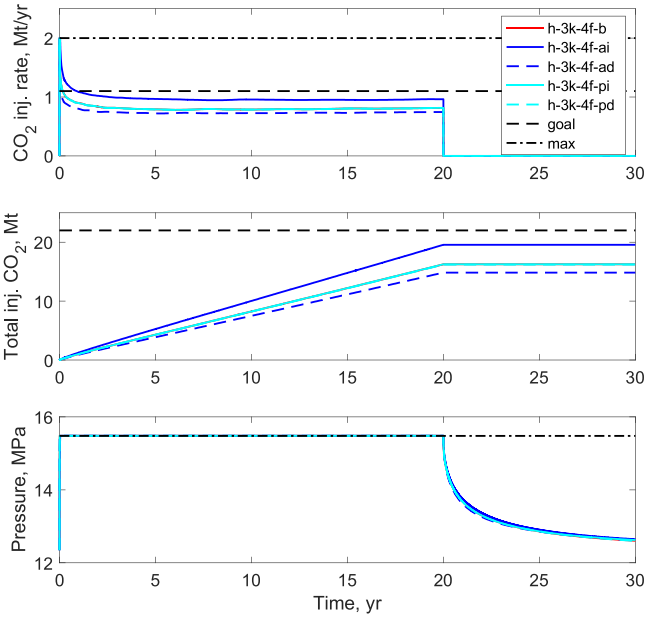


Fig. 8. Plot showing CO_2 injection rate, total CO_2 injected, and pressure at the injection well for horizontal wells with different fracture-related areas. The simulations are compared against the maximum allowable (rate and pressure) and optimum (rate and total amount) value. Note that the base case (case-h-3k-4f-b) and the case with decreased permeability (case-h-3k-4f-pd) are plotted under the solid cyan line representing the case with increased permeability (case-h-3k-4f-pi).

Table 3
Volume where free phase CO_2 is above a saturation of 0.01 and depth plus associated area where CO_2 has the largest lateral area at 30 years.

	Volume, km^3	Area, km^2	Depth, m
Case-v	0.404	8.9	−1,041.0
Case-h-3k	0.408	12.7	−1,041.0
Case-h-3k-4f-b	0.510	10.5	−1,041.0
Case-h-6k	0.482	13.2	−1,041.0
Case-h-6k-8f	0.617	12.7	−1,041.0
Case-h-6k-8f-ai-pi	0.961	45.6	−959.9

fluid if the plume intersected a conduit. It has implications for permitting of the operation and assessing the stage of the operation as a project moves towards site closure. Further, it is expected that the operation will stay within an area where the pore space is owned or controlled by the operator, thus a larger CO_2 footprint will require more pore space ownership. Table 3 presents the lateral area where CO_2 saturation is present above 0.01 vol fraction. These areas are also shown in the Fig. 9. As expected, the vertical well has the smallest footprint and the footprint grows incrementally for the horizontal wells. This trend matches the amount of CO_2 injected. Surprisingly, the presence of the hydraulic fractures (and increase in total CO_2 injected) result in a slight decrease in the plume area. This is likely due to more CO_2 being distributed vertically from the fracture face. The largest area change was during Case-h-6k-8f-ai-pi, which saw a greater than five times increase in area versus case-v. The depth (-959.9 m) indicates that the thin but high permeability streak in the Elmhurst is the source of this large area increase. This increase in CO_2 plume area must be balanced with the fact that this was the only scenario that was able to achieve the original storage objective.

3.1.2. Disposition of the CO_2 over time

A critical component of long-term risk assessment is determining the disposition of CO_2 in the subsurface. In its riskiest form, CO_2 exists as a buoyant free phase supercritical fluid (or gas) that could intersect a

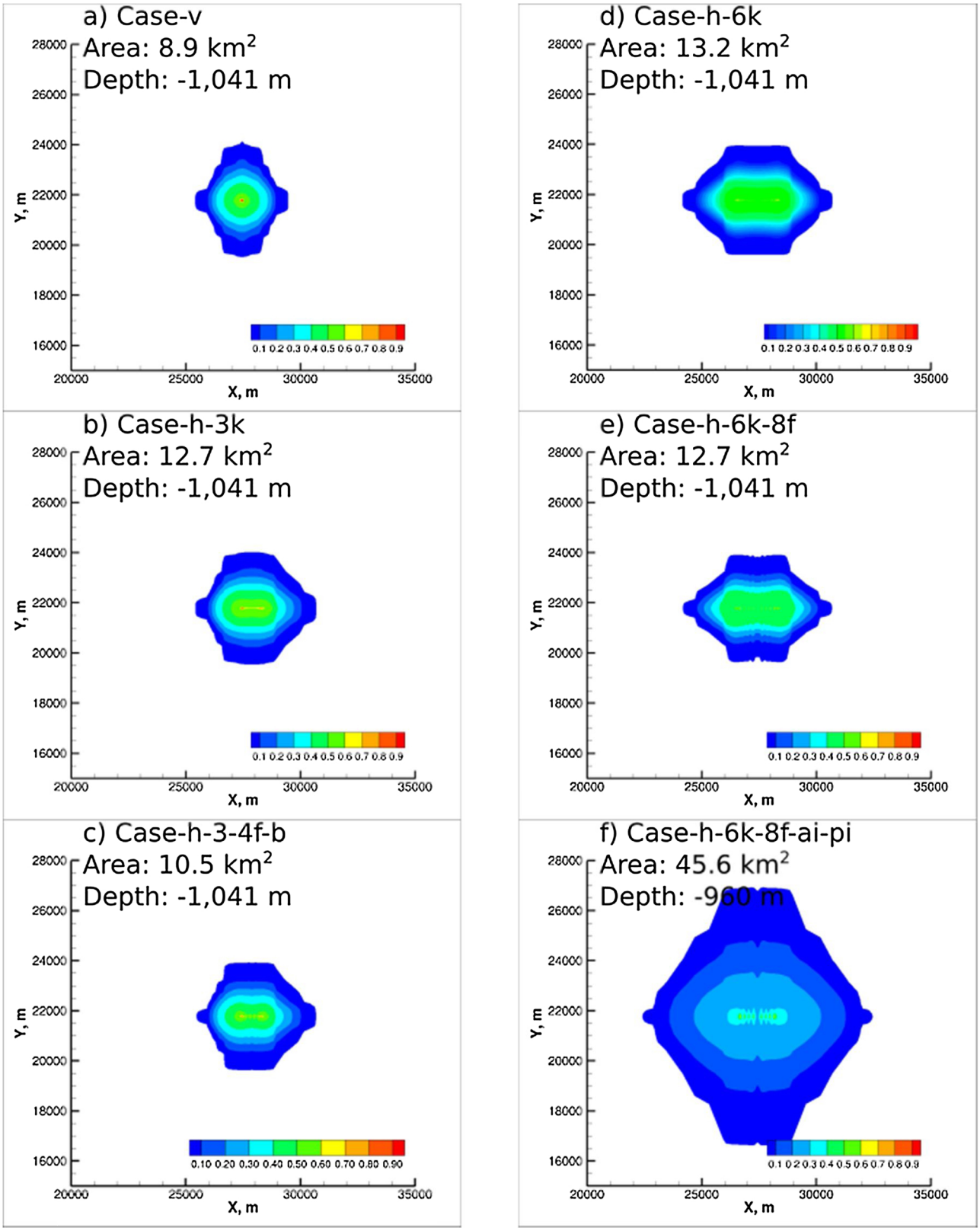


Fig. 9. Map view of CO_2 saturation along the layer of maximum areal extent at 30 years for the cases presented in Table 3.

conductive pathway (e.g., a fault or compromised well) and leak into overlying sensitive formations or leak back into the atmosphere. However, it is expected that over time CO_2 will be permanently stored as dissolved CO_2 in the formation brine, as a free but discontinuous and immobile phase (e.g., trapped via capillary trapping), or consumed as part of secondary mineral precipitation. The latter mechanism is a long-term process that is out of the scope of this work, but we can look at the evolution of CO_2 from free phase to trapped and dissolved to characterize how risk evolves for the different cases. Table 4 shows, for select cases, the total mass of CO_2 in the system and by percentage of

Table 4
Total mass and percent of CO_2 into free phase (FP), aqueous (AQ), and trapped phase (TP) at injection end (20 yr) and simulation end (30 yr) for several cases.

		20 years			30 years		
	Total, Mt	FP, %	AQ, %	TP, %	FP, %	AQ, %	TP, %
Case-v	13.14	84.0	16.0	0.0	78.7	17.4	3.8
Case-h-3k	14.81	84.2	15.9	0.0	79.3	17.0	3.8
Case-h-3k-4f-b	16.25	83.2	16.8	0.0	77.9	18.2	3.8
Case-h-3k-4f-ai	19.56	78.1	21.9	0.0	73.2	23.7	3.1
Case-h-3k-4f-ad	14.84	84.2	15.8	0.0	79.3	17.0	3.8
Case-h-3k-4f-pi	16.25	83.2	16.9	0.0	77.9	18.2	3.8
Case-h-3k-4f-pd	16.19	83.2	16.8	0.0	78.0	18.2	3.8
Case-h-6k	17.54	83.9	16.1	0.0	78.7	17.4	3.9
Case-h-6k-8f	19.10	82.7	17.3	0.0	77.3	18.8	3.9
Case-h-6k-8f-ai-pi	22.44	77.7	22.3	0.0	72.7	24.1	3.2

the total mass: the distribution of free phase (FP) CO_2, CO_2 dissolved in brine (AP), and CO_2 that is trapped (TP) in pore spaces. In all cases, mobile free phase CO_2 decreases by an average of 5.2 % (standard deviation 0.20) as CO_2 in the aqueous phase and trapped CO_2 increases from injection stop to simulation end. This decrease in mobile CO_2 shows no strong correlation to well orientation, fracture number, or fracture properties. As expected, no scenario shows a generally lower (or higher) risk profile over time.

There are, however, correlations with the initial distribution of CO_2 and the different parameters studied in the various cases. The vertical versus horizonal well orientation cases (without fractures) do not show a significant difference in the distribution of immobilized CO_2 (the combined trapped and aqueous phase amounts) over time. The vertical well is 21.3 %, the 3k horizontal well is 20.8 %, and the 6k horizontal well is 21.3 %. However, when the horizontal wells with no fractures are compared to wells that have been hydraulically fractured there is a small but systematic increase in the immobilized phase for the 3k well (1.32 %) and the 6k well (1.39 %). When fracture area is increased, there is an impact to the immobilized phase for the 3k wells (4.76 % increase) and 6k wells (6% increase). The modest increase is due to the presence of the fracture area, which provides an increase in the amount of CO_2 that can dissolve into the aqueous phase, though it has less impact on the trapped CO_2.

This initial CO_2 distribution does play a role in the initial risk magnitude and time to achieve acceptable risk for site closure but generally, there is no significant positive or negative impact from hydraulic fracturing on the disposition of CO_2 during the time simulated.

3.2. Performance and uncertainty

3.2.1. Injectivity and capacity: did we achieve our performance goal

The results show that horizontal wells do improve injectivity and capacity and that there is an additional modest increase when the well is made longer and with more fractures. The real benefit, in this scenario at least, is linking up the high permeability intervals so that the operator can maximize injection into the laterally continuous high permeability zones.

3.2.2. Uncertainty and the things accounted for and not accounted for in our work

Geochemical response and physical alteration of the fractures was not studied in this work, but it does bear mentioning the impact that some of these phenomena would have on the results. Numerous laboratory- and simulation-based studies have been performed to evaluate the impact of CO_2 and coinjected gases, such as H_2S and SO_2, on fluid flow in storage reservoirs (Cantucci et al., 2015; Izgec et al., 2008a, b; Xu et al., 2007). These efforts have revealed that mineral alterations during CO_2 injection can result in significant changes in porosity and permeability within CO_2 storage reservoirs. Modeling and experimental studies indicate that porosity can increase by as much as 43 % in acidified zones near the injection point, and decreases of as much as 23 % can occur in the higher pH peripheral zones where precipitation can occur. Additionally, these studies found that incongruent mineral dissolution tended to increase surface roughness and led to reductions in CO_2 flow rates and pressure increases within the storage reservoir, leading to negative overall impacts on reservoir performance.

3.3. Worth of hydraulic fracturing in geologic carbon storage

Based on our analyses, we conclude that the CCUS community can build upon decades of experience manipulating the subsurface for fossil fuel production to safely engineer geologic carbon storage reservoirs to meet the operational needs for an economically-viable technology to reduce CO_2 emissions. While hydraulic fracturing will not be appropriate for all applications, it should be considered as an approach to improve the performance of low permeability reservoirs. As identified previously, it will be important to understand the stress conditions at prospective storage sites, as compressional stress regimes are favorable for limiting injection-induced mechanical failure of overlying caprocks (Rutqvist et al., 2008). However, as demonstrated by the large volume of anecdotal data available from the oil and gas industry (EIA, 2018), reservoir hydraulic fracturing can be performed with minimal risk of impacting overlying underground sources of drinking water.

4. Conclusions

A prior modeling study was conducted to evaluate the impact of hydraulic fracturing for increasing injectivity and capacity of CO_2 storage reservoirs (Ghaderi et al., 2009). Their results indicated that fracturing can increase injectivity and storage capacity by modest to significant amounts (10%–35%). Our simulations support these earlier findings, and demonstrated that horizontal wells with a series of hydraulically-induced fractures maximized reservoir performance (both injectivity and storage) over nonfractured horizontal and vertical injection wells. These findings highlight that targeted applications of hydraulic fracturing could be used to increase the performance of low permeability reservoirs. In particular, hydraulic fracturing could be used in a horizontal well to connect two or more high permeability zones that are separated by an impermeable or low-permeability baffle. This would allow a single well to broadly access both layers, thereby maximizing total storage capacity of the reservoir, which is an important consideration for addressing the large volumes of anthropogenic CO_2 that must be injected to meet global atmospheric CO_2 reduction goals (Paris Agreement, 2015). To further develop this as an option for CCUS additional work is needed in model verification using pilot-scale field data, additional risk assessment, in public outreach and gaining public acceptance, and in quantifying the economic feasibility through a cost-benefit analysis.

CRediT authorship contribution statement

Nicolas J. Huerta: Conceptualization, Methodology, Software, Investigation, Writing - original draft, Writing - review & editing. **Kirk J. Cantrell:** Conceptualization, Writing - original draft. **Signe K. White:** Conceptualization, Methodology, Software. **Christopher F. Brown:** Conceptualization, Writing - review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.ijggc.2020.103105.

References

Adachi, J., Siebrits, E., Peirce, A., Desroches, J., 2007. Computer simulation of hydraulic fractures. Int. J. Rock Mech. Min. Sci. 44 (5), 739–757.

Appriou, D., 2019. Assessment of the Geomechanical Risks Associated With CO_2 Injection at the FutureGen 2.0 Site. Application of the State of Stress Assessment Tool (SOSAT). PNNL-28657.

Battelle, 2011. Appalachian Basin – R.E. Burger Plant Geologic CO_2 Sequestration Field Test. Battelle, Columbus, OH.

Bear, J., 1972. Dynamics of Fluids in Porous Media. American Elsevier Publishing Company, Inc., New York.

Cantucci, B., Montegrossi, G., Buttinelli, M., Vaselli, O., Scrocca, D., Quattrocchi, F., 2015. Geochemical barriers in CO_2 capture and storage feasibility studies. Transp. Porous Media 106, 107–143.

DOE, 2015. Carbon Storage Atlas, 5th edition. .

EIA, 2018. Hydraulically Fractured Horizontal Wells Account for Most New Oil and Natural Gas Wells. URL: https://www.eia.gov/todayinenergy/detail.php?id=37815#. Date accessed: 03/04/2020. .

EPA, 2010. Federal requirements under the underground injection control (UIC) program for carbon dioxide (CO_2) geologic sequestration (GS) wells. Fed. Regist. 75 (237).

EPA, 2013a. Geologic Sequestration of Carbon Dioxide, Underground Injection Control (UIC) Program Class VI Well Site Characterization Guidance. EPA 816-R-13-004.

EPA, 2013b. Geologic Sequestration of Carbon Dioxide, Underground Injection Control (UIC) Program Class VI Well Area of Review Evaluation and Corrective Action Guidance. EPA 816-R-13-005.

FutureGen Industrial Alliance, 2013. Underground Injection Control Permit Applications for FutureGen 2.0 Morgan County Class VI UIC Wells 1, 2, 3, and 4 – Supporting Documentation, Jacksonville, Illinois.

FutureGen Industrial Alliance, 2015. Scientific and Technical Report. Pipeline and Regional Carbon Capture Storage Project. DE-FE0001882.

Ghaderi, S., Keith, D., Leonenko, Y., 2009. Feasibility of injecting large volume of CO_2 into aquifers. Energy Procedia 1 (1), 3113–3120.

Gilmore, T., Bonneville, A., Sullivan, C., Kelley, M., Appriou, D., Vermeul, V., White, S., Zhang, Z., Bjornstad, B., Cornet, F., Gerst, J., Gupta, N., Hund, G., Horner, J., Last, G., Lanigan, D., Oostrom, M., McNeil, C., Moody, M., Rockhold, M., Elliott, M., Spane, F., Strickland, C., Swartz, L., Thorne, P., Brown, C., Hoffman, J., Humphreys, K., 2016. Characterization and design of the FutureGen 2.0 carbon storage site. Int. J. Greenh. Gas Control. 53, 1–10.

Greenberg, S., Bauer, R., Will, R., Locke, I.I.R., Carney, M., Leetaru, H., Medler, J., 2017. Geologic Carbon storage at a one million tonne demonstration project: lessons learned from the Illinois Basin – Decatur Project. Energy Procedia 114, 5529–5539.

Izgec, O., Demiral, B., Bertin, H., Akin, S., 2008a. CO_2Injection into Saline Carbonate Aquifer Formations I: Laboratory Investigation. Transp. Porous Media 72, 1–24.

Izgec, O., Demiral, B., Bertin, H., Akin, S., 2008b. CO_2 injection into saline carbonate aquifer formations II: comparison of numerical simulations to experiments. Transp. Porous Media 73, 57–74.

Kell, S., 2011. State Oil and Gas Agency Groundwater Investigation and Their Role in Advancing Regulatory Reforms. A Two-State Review, Ohio and Texas.

Kelly, M., Moody, M., Zeller, E., Rike, W., Berelsman, N., McNeil, C., Holley, J., Sullivan, C., Appriou, D., Spane, F., Horner, J., Gilmore, T., 2012. Borehole Completion and Characterization Summary Report for the Stratigraphic Well, Morgan County, Illinois. FG-RPT-015 Rev 1/PNWD-4343. FutureGen Industrial Alliance, Washington, DC.

McGrail, B., Spane, F., Sullivan, E., Bacon, D., Hund, G., 2011. The Wallula basalt sequestration pilot project. Energy Procedia 4 (2011), 5653–5660.

McGrail, B., Spane, F., Amonette, J., Thompson, C., Brown, C., 2014. Injection monitoring at the Wallula basalt pilot project. Energy Procedia 63 (2014), 2939–2948.

Nghiem, L., Sammon, P., Grabenstetter, J., Ohkuma, H., 2004. Modeling CO_2 storage in aquifers with a fully-coupled geochemical EOS compositional simulator. In: SPE 89474, Presented at the SPE/DOE Symposium on Improved Oil Recovery. Tulsa. April 17-21.

Pan, P., Rutqvist, J., Feng, X., Yan, F., 2013. Modeling of caprock discontinuous fracturing during CO_2 injection into a deep brine aquifer. Int. J. Greenh. Gas Control. 19, 559–575.

Paris Agreement, 2015. United Nations Treaty Collection. December 12. Registered on November 4, 2016. Retrieved September 13, 2019.

Pruess, K., Garcia, J., Kovscek, T., Oldenburg, O., Rutqvist, J., Steefel, C., Xu, T., 2004. Code intercomparison builds confidence in numerical simulation models for geologic disposal of CO_2. Energy 29 (9–10), 1431–1444.

Raziperichikolaee, S., Alvarado, V., Yin, S., 2013. Effect of hydraulic fracturing on long-term storage of CO_2 in stimulated saline aquifers. Appl. Energy 102, 1091–1104.

Rutqvist, J., Birkholzer, J., Chin-Fu, T., 2008. Coupled reservoir-geomechanical analysis of the potential for tensile and shear failure associated with CO_2 injection in multi-layered reservoir-caprock systems. Int. J. Rock Mech. Min. Sci. 45 (2), 132–143.

Snow, D., 1965. A Parallel Plate Model of Fractured Permeable Media. Ph.D. Thesis. University of California at Berkeley.

Sullivan, E., Horner, J., Appriou, D., Brown, C., Gilmore, T., 2015. Continental sequence stratigraphy of the Mount Simon at the FutureGen 2.0 site in Morgan County, Illinois. In: PNWD-SA-10484. Presented at the 2015 Carbon Capture, Utilization, and Sequestration Conference. Pittsburgh, PA.

White, M., Bacon, D., McGrail, B., Watson, D., White, S., Zhang, Z., 2012. STOMP Subsurface Transport Over Multiple Phases: STOMP-CO2 and STOMP-CO2E Guide: Version 1.0. PNNL-21268. Pacific Northwest National Laboratory, Richland, WA.

White, S., Zhang, Z., Oostrom, M., 2016. Simulation of carbon dioxide injection at the FutureGen2.0 site: class VI permit model and local sensitivity analysis. Int. J. Greenh. Gas Control. 55, 177–194.

Xu, T., Apps, J., Pruess, K., Yamamoto, H., 2007. Numerical modeling of injection and mineral trapping of CO_2 with H_2S and SO_2 in a sandstone formation. Chem. Geol. 242, 319–346.

Yonkofski, C., Tartakovsky, G., Huerta, N., Wentworth, A., 2019. Risk-based monitoring designs for detecting CO_2 leakage through abandoned wellbores: an application of NRAP's WLAT and DREAM tools. Int. J. Greenh. Gas Control. 91.

Zhang, L., Dilmore, R., Namhata, A., Bromhal, G., 2018. Feasibility of CO_2 migration detection using pressure and CO_2 saturation monitoring above an imperfect primary seal of a geologic CO_2 storage formation: a numerical investigation. Comput. Geosci. 22 (3), 909–923.